

Video Optimization

Performance Report

Key Results

91% Size Reduction • 8.5s → 0.18s Loading Time • <0.5s First Frame

29.67 MB → 2.67 MB • 6 Videos Optimized • Per-Video Posters Added

April 2026 • Technical Analysis

The Problem

Hero section videos were loading slowly, causing delayed first-frame display and poor user experience. Videos were served with excessive file sizes and non-optimized playback settings.

Impact Timeline:

User sees blank screen until:

3-8 seconds

(on 4G connection)

6 Root Causes Identified

Excessive Bitrates

Up to 17.6 Mbps (Blu-ray quality)

Unneeded Audio Tracks

Muted videos with AAC audio

No Faststart

Moov atom at end of file

preload="none"

Zero buffering before play()

Generic Poster

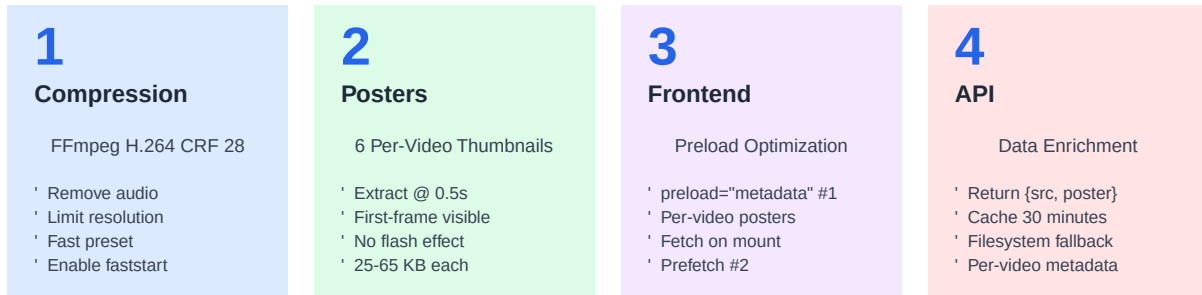
Same image for all videos

No API Poster Info

Frontend can't select per-video

Solution Strategy

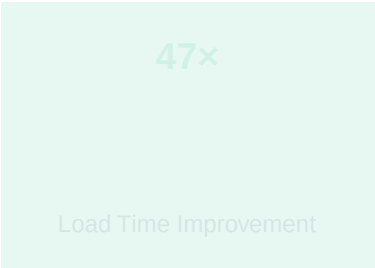
A 4-layer optimization approach addressing file size, streaming capability, and perceived performance:



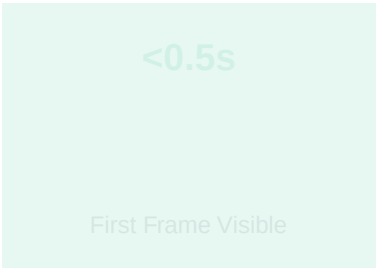
Optimization Pipeline Flow



Results

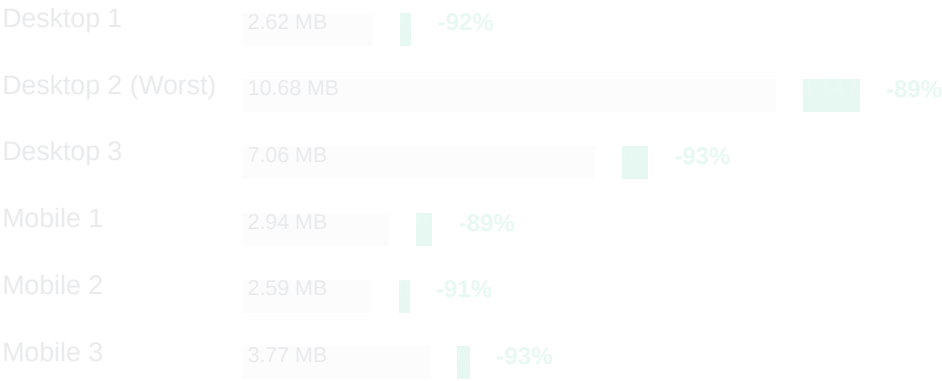


faster



vs 3-8s

Per-Video Compression Results



Technical Implementation

FFmpeg Compression Command

```
ffmpeg -i Desktop1.mp4 -vcodec libx264 -crf 28 -preset slow \
-vf "scale='min(iw,1920)':-2" -movflags +faststart -an -y Desktop1_compressed.mp4
```

Key Parameters

libx264	H.264 codec - 99%+ browser compatible
CRF 28	Quality factor (0=lossless, 51=worst) - optimal web quality
preset slow	Slower encoding but better compression ratio
scale 1920x?	Max width without upscaling, height auto
+faststart	Move metadata to file start for progressive streaming
-an	Remove audio tracks (muted video anyway)

Frontend Changes

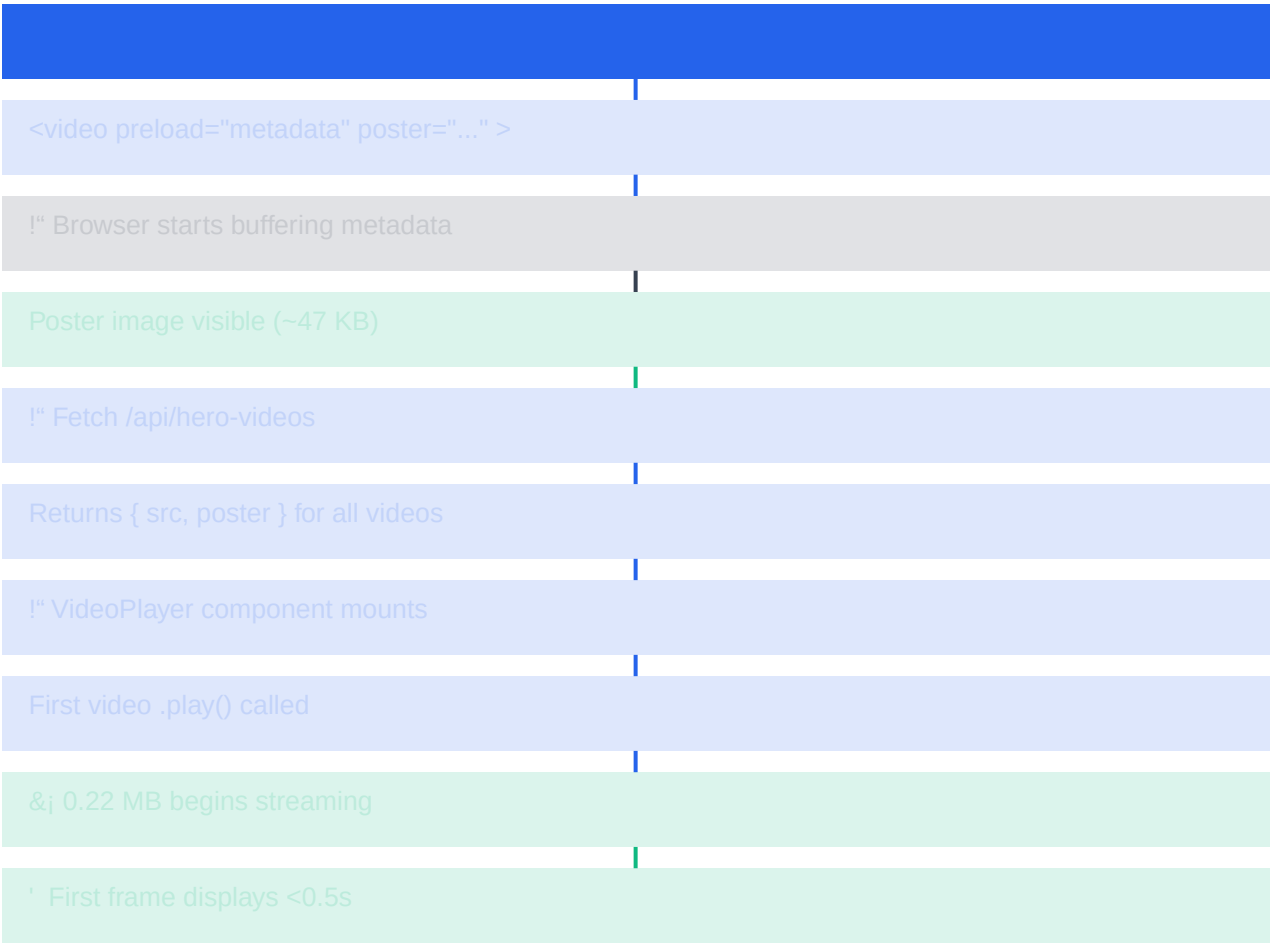
- **HeroSection.tsx**
Type VideoEntry { src, poster }, preload="metadata" on #1
- **hero-videos/route.ts**
Returns { src, poster } objects, scans for *_poster.jpg
- **videos/*.mp4**
Replaced with compressed versions
- **videos/*_poster.jpg**
6 new per-video thumbnail images

Backend Changes

- **VideoController.php**
Hero endpoint returns { src, poster } objects with Storage import

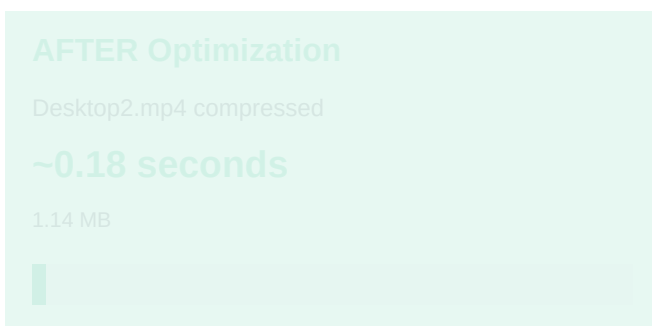
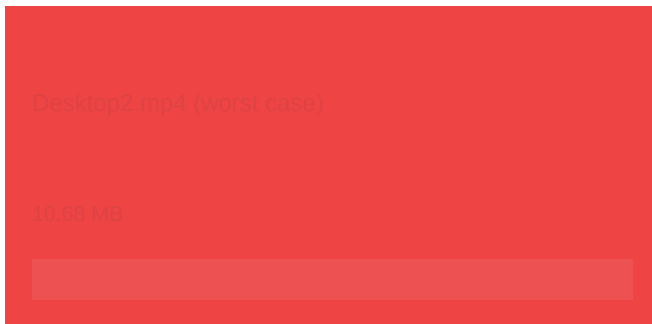
Complete Data Flow (After Optimization)

User opens homepage



Performance Before vs After

Load Time Comparison (4G Network - 10 Mbps)



&j 47× Faster Loading Speed

Desktop2 reduced from 8.5 seconds to 0.18 seconds. On 3G (1.5 Mbps), improvement is from ~57s to ~1.2s.

Key Improvements

File Size

Before:
29.67 MB

After:
2.67 MB

91% smaller

Load Time

Before:
~8.5 seconds

After:
~0.18 seconds

47× faster

First Frame

Before:
3-8 seconds

After:
<0.5 seconds

Near instant

User Experience

Before:
Blank screen

After:
Poster visible

No flash effect

Audio Overhead

Before:
~150 KB/video

After:
Removed

No benefit

Mobile Impact

Before:
3+ seconds

After:
<200ms

Smooth load

Future Recommendations

- 1 WebM/VP9 Format**
Convert to next-gen codecs for 20-30% additional compression on modern browsers, with MP4 fallback
Impact: **High**
Effort: **Medium**
- 2 Adaptive Bitrate (HLS/DASH)**
Serve multiple quality tiers based on detected bandwidth for optimal streaming
Impact: **High**
Effort: **High**
- 3 CDN Deployment**
Host videos on CDN (CloudFront, Cloudflare) for reduced latency and geo-distributed delivery
Impact: **Very High**
Effort: **Medium**
- 4 Auto-Compress on Upload**
The backend already has CompressVideoJob - ensure all future uploads go through it
Impact: **Medium**
Effort: **Low**

Summary

1. Video file sizes reduced by 91% through optimized FFmpeg compression (H.264 CRF 28, faststart)
2. First visible frame time cut from 3-8 seconds to <0.5 seconds using per-video poster thumbnails
3. API enhanced to return { src, poster } per video, enabling frontend-level optimization
4. Frontend updated with intelligent preload strategy (metadata on first video, none on rest)
5. Zero UI/UX changes - improvements are invisible to users but dramatically improve perceived performance

Impact on User Experience:

- ' **Faster page load times - reduces bounce rate**
- ' **Better SEO - Core Web Vitals improve (LCP optimized)**
- ' **Lower bandwidth costs - 91% reduction in data transfer**
- ' **Improved mobile experience - critical for 4G/3G users**
- ' **Professional perception - fast-loading sites feel more trustworthy**